Micro-Robot Swarm

Project plan

M1–M5 delivery and planned expansion through M8

Desktop supervision of ESP32-C6 robots
using external visual feedback

Project objective and system scope

An operator draws a route. A robot follows it using camera feedback.

Desktop application	Camera acquisition, calibrated localization, planning and supervisory motion control
Robot firmware	Motor regulation, command watchdog and local safe stop on each ESP32-C6

C++20 / Qt 6 / OpenCV BLE commands
macOS development and early Linux support

Application architecture

Camera and vision	Timestamped frames feed AprilTag detection and calibration. Live cameras and recorded video share one interface.
Tracking and world	The tracker filters observations and marks stale poses. One authoritative WorldModel holds metric robot state.
Planning and control	Tasks produce paths and motion references. Supervisory control compares desired motion with fresh visual feedback.
Robot communication	RobotManager owns connections. A transport interface isolates BLE from planning and control.

The GUI observes services and accepts operator commands. Worker services keep camera and control processing responsive.

M1–M5 capability plan

M1	**Foundation**	Responsive camera preview, replay/simulation and portable build/test infrastructure
M2	**Localization**	AprilTags, camera/table calibration and fresh robot poses in metres
M3	**Communication**	BLE command/telemetry, manual control and emergency stop
M4	**Point control**	Vision feedback drives one robot to a selected target with safe stop rules
M5	**Path tracking**	Mouse-drawn paths, smoothing, trajectory generation and Pure Pursuit

M1–M2: camera foundation and localization

M1: reliable input	Select, start and stop a camera. Handle removal safely and use recorded video for repeatable development.
M1: core services	Separate camera and transport adapters from the core. Add configuration, logging and hardware-free GoogleTest/CTest checks.
M2: calibration	Estimate camera intrinsics and distortion. Calibrate the tabletop transform and save profiles per camera/resolution.
M2: robot state	Associate AprilTags with robot IDs. Produce filtered metric poses, with timestamps and fresh/stale/lost states.

Exit evidence: responsive video and measured position/orientation accuracy

M3–M4: communication and feedback control

M3: BLE interface	Discover and connect an ESP32-C6. Exchange versioned commands and telemetry with sequence and freshness checks.
M3: safe operation	Provide manual control and emergency stop. Verify reconnect behaviour and the robot-side command watchdog.
M4: target control	Select a target in world coordinates. Use a differential-drive model to generate bounded linear and angular velocity commands.
M4: safety states	Pause or stop when tracking becomes stale or communication fails. Record commands and measured pose for evaluation.

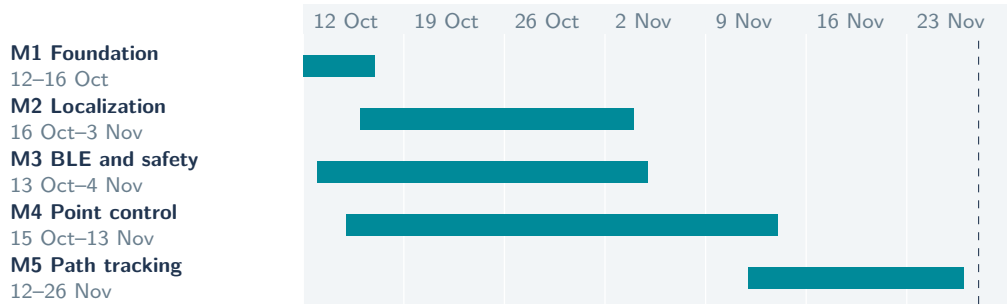
M3 builds on M1. Integrated M4 control requires both M2 localization and M3 communication.

M5: operator paths and trajectory tracking

Operator input	Select a robot and draw a route on the calibrated view. Store the path in metres so display resizing preserves its geometry.
Path preparation	Simplify and smooth the drawn points while respecting workspace boundaries. Generate a bounded motion reference.
Path following	Use Pure Pursuit as the baseline controller. Show the desired path, actual robot trace and execution progress.
Execution and tests	Support pause, cancel and safe completion. Evaluate representative shapes using cross-track RMSE, maximum error and completion time.

M5 completes the core interaction after M4 point control is validated

M1–M5 Gantt chart



Target: 28 November 2026

Baseline snapshot: 10 Oct 2026. Later Linux additions require schedule review. Bars show plans.

Planned expansion through M8

M6	**Autonomous paths**	Manual obstacles, occupancy grid, footprint inflation and A* for one robot
M7	**Coordination**	Concurrent robot tasks, conflict prediction and prioritized or reservation planning
M8	**Portability**	Windows build/deployment and device verification, with adapters that preserve the core

Optional extensions: Zigbee, second-camera fusion, onboard Wi-Fi video, LiDAR and task allocation

Expansion dates remain to be planned after core validation

Validation and project completion

M1–M5 plus quantitative evaluation form the minimum thesis system

Capability checks	Demonstrate calibrated tracking, BLE control and repeatable path execution
Safety checks	Verify stale-pose stop, lost-connection response, watchdog and emergency stop
Thesis evidence	Report localization error, latency, path RMSE/max error and completion time